

Deep Learning Portfolio Examination

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Part 3: Working with the New, Small “organs” Dataset

Overview

The new organs dataset is very small. It has only 450 training images spread across 11 classes, with 200 more images kept aside for testing. When you have this little data, training a model from zero usually does not work well, because the model simply does not see enough examples to learn the classes properly. So instead of starting from nothing, we reused what our models had already learned from the larger orgs dataset. This idea is called transfer learning. Because orgs and organs are both grayscale images with the same 11 classes, the knowledge carries over very cleanly. As slimmed models were beating the original models in terms of accuracy so we selected the trained models of slimmed for the transfer learning approaches.

The Three Approaches We Tested

1. From scratch : Train the model from random starting weights using only the organs data, with no transfer at all. We ran this for both the original architecture and the slimmed architecture. So the first table below is the true baseline and the second table (slimmed models) shows how the downscaling from Part 2 affects a model trained from scratch on this small dataset.

2. Feature extraction : Take a model already trained on orgs, keep all of its learned feature layers frozen, and only train the final classification layer. The model reuses everything it learned before and just learns how to match those features to the organ classes. This is the approach recommended for a small dataset from a similar domain.

3. Fine-tuning : Take the same pretrained model, but this time let all the layers keep learning, using a smaller learning rate so we do not wash away what the model already knew.

Results

The tables below show the evaluation metrics accuracy, precision, recall, and macro F1 for each model on the organs test set. From scratch results are given for both the original and the slimmed models, followed by the two transfer learning approaches.

1a. From scratch — original models

Model	Accuracy (%)	Precision	Recall	Macro F1
AlexNet	32.0	0.172	0.261	0.171
VGG16	54.0	0.445	0.442	0.390
ResNet18	50.5	0.459	0.447	0.413

1b. From scratch — slimmed models

Model	Accuracy (%)	Precision	Recall	Macro F1
AlexNet	43.0	0.220	0.348	0.259
VGG16	56.5	0.507	0.513	0.480
ResNet18	60.5	0.646	0.546	0.525

2. Feature extraction (freeze features, train head) on orgs trained models(slimmed)

Model	Accuracy (%)	Precision	Recall	Macro F1
AlexNet	61.5	0.577	0.534	0.547

VGG16	61.0	0.561	0.553	0.548
ResNet18	61.5	0.542	0.538	0.535

3. Fine-tuning (train all layers, small learning rate) on orgs trained models(slimmed)

Model	Accuracy (%)	Precision	Recall	Macro F1
AlexNet	56.0	0.507	0.462	0.457
VGG16	67.5	0.615	0.625	0.615
ResNet18	63.5	0.575	0.565	0.548

What the Results Tell Us

Transfer learning clearly beat training from scratch for every model : The biggest jump was AlexNet, which went from 43.0% up to 61.5% with feature extraction. VGG16 and ResNet18 also improved. This confirms that reusing the features learned on orgs is much better than starting from nothing on such a small dataset.

Even without transfer, the slimmed models beat the original ones : Trained from scratch, every slimmed model scored higher than its original version for example ResNet18 60.5% vs 50.5%, and AlexNet 43.0% vs only 32.0%. On such a small dataset the smaller models overfit less, so downscaling helped here too. Note the original AlexNet did not even reach the 40% target, while the slimmed one did.

Feature extraction was steady and safe : All three models landed close together, around 61–62%. It is the dependable option you get a solid result no matter which model you pick.

Fine-tuning gave the best single result, but only for the bigger models: VGG16 fine-tuned reached 67.5%, the highest score in the whole experiment, and ResNet18 also improved to 63.5%. These larger models have richer features, so letting them adapt to organs helped.

Fine-tuning hurt the smallest model. AlexNet actually dropped to 56.0% when fully fine-tuned.

Recommendations

For the organs data as it stands today, fine-tuned VGG16 is the best performer (67.5%) and is our top recommendation if raw accuracy is the priority.

If we want a safe, reliable option that will not overfit especially for smaller models feature extraction is the better choice. It gave consistent results with no overfitting, which matters a lot when data is scarce.

As more organs data is collected in the future, fine-tuning will become more and more attractive, because more data reduces the overfitting risk and lets the bigger models adapt fully. For now, with only 450 images, we recommend feature extraction as the default and fine-tuning only for the larger backbones (VGG16 and ResNet18), where it clearly helps.